MILITARY SPECIFICATION

LUBRICATING AND SERVICING UNIT, POWER OPERATED: SKID-MOUNTED

1. SCOPE

1.1 This specification covers one type of power-operated, skid-mounted lubricating and servicing unit.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

GG-G-91	-	Gage, Tire Pressure, Self-Contained And Inflator Gage, Pneumatic Tire.
QQ-A-200/9	-	Aluminum Alloy Bar, Rod, Shapes, Tube, And Wire, Extruded, 6063.
QQ-S-700	-	Steel Sheet And Strip, Medium And High Carbon.
QQ-S-741	-	Steel, Carbon Structural Shape, Plate And Bar.
TT-P-664	-	Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-resisting.
GGG-B-451	-	Blade, Hand Hacksaw And Blade, Power Hacksaw.
GGG-F-671	-	Frame, Hand Hacksaw, Frame, Jeweler S Saw, And Frame, Coping Saw.
GGG-G-770	-	Gun, Air Blow, (Blast Cleaning).
GGG-O-591	-	Oiler, Hand.
PPP-B-636	-	Box, Fiberboard.
PPP-D-705	-	Drum, Metal Shipping, Steel, (Over 12 And Under 55 Gallon).
PPP-D-729	-	Drum, Metal, 55-Gallon (For Shipment Of Noncorrosive Material).

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PPP-P-40	-	Packaging And Packing Of Hand Tools.
PPP-T-60	-	Tape, Pressure-Sensitive Adhesive. Waterproof For Packaging.

MILITARY

MIL-C-104	-	Crate, Wood, Lumber And Plywood Sheathed, Nailed And Bolted.
MIL-P-116	-	Preservation, Methods Of.
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	-	Varnish, Moisture And Fungus Resistant (For Treatment Of Communications, Electronic, And Associated Equipment.
MIL-S-207	-	Sulfuric Acid, Electrolyte, Packaging, Packing, And Marking For Shipment And Storage Of.
MIL-P-514	-	Plate, Identification, Instruction And Marking, Blank.
MIL-T-704	-	Treatment And Painting Of Materiel.
MIL-F-3541	-	Fittings, Lubrication.
MIL-C-3600	-	Compressor, Air And Gas (Except Oxygen) Packaging Of.
MIL-C-3774	-	Crates, Wood, Open, 12,000-And-16,000.
MIL-G-3859	-	Grease Gun, Hand, High Pressure, Lever-operated, Cartridge And Bulk Loading (14 ounce and 21 ounce capacity).
MIL-H-3868	-	Hose Assemblies, Grease Gun, High And Low Pressure.
MIL-C-4109	-	Coupling Assembly, Low Pressure, Air Hose, Quick-disconnect.
MIL-L-4387	-	Lubrication Dispensing Equipment Accessories.
MIL-E-10062	-	Engine, Preparation For Shipment And Storage Of.
MIL-M-10304	-	Meter, Electrical Indicating, Panel Type, Ruggedized General Specification For.
MIL-G-10924	-	Grease, Automotive And Artillery.
MIL-E-11275	-	Engine, Gasoline, Industrial Type, General Specification For.
MIL-C-13486	-	Cable, Special Purpose, Electrical, Low-tension Heavy Duty Single-conductor And Multiconductor.

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MIL-E-16298	-	Electric Machines Having Rotating Parts And Associated Repair Parts, Packaging Of.
MIL-G-23827	-	Grease, Aircraft And Instrument, Gear And Actuator Screw.
MIL-E-46717	-	Engine, Gasoline, Air Cooled, Industrial Type, 4 Cycle, Military Design, 1-1/2, 3 And 6 Net Continuous Horsepower.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-129	-	Marking For Shipment And Storage.
MIL-STD-705	-	Generator Sets, Engine-driven, Methods Of Tests And Instructions.
MS24538	-	Transmitter, Pressure 24 Volts DC 60 PSI.
MS24541	-	Indicator, Pressure 24 Volts DC, Range 0-60 PSI.
MS24544	-	Indicator, Liquid Quantity, 24 Volts, DC, Engine Fuel.
MS35000	-	Battery, Storage, Lead-Acid, Waterproof.
MS35392	-	Inner Tube, Pneumatic Tire.
MS35876	-	Generator, Engine Accessory, D.C. 28-volt, 5-Ampere.
MS49006	-	Plug, Pipe, Magnetic-headless, Aluminum, Iron Or Steel.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Tariff Order Section, 1616 P Street, N. W., Washington, D. C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook - Recommended Practices

(Application for copies should be addressed to the Society of Automotive Engineers, Two Pennsylvania Plaza, New York, N. Y., 10001.)

RUBBER MANUFACTURERS ASSOCIATION

Engineering Standards for Multiple V-Belt Drives

(Application for copies should be addressed to the Rubber Manufacturers Association, 444 Madison Avenue, New York, N. Y., 10022.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code: Section VIII - Unfired Pressure Vessels, Section IX-Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, N. Y., 10017.)

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AMERICAN WELDING SOCIETY

Standard Qualification Procedure B-30

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, N. Y., 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A268 - Seamless Welded Ferrous Stainless Steel Tubing for General Service

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19102.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standard for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 First article. The supplier shall furnish a first article for examination and test within the time frame specified (see 6.2) to prove, prior to starting production, that his production methods and choice of design detail will produce units which comply with the requirements of this specification. Examination and tests shall be as specified herein and shall be subject to surveillance and approval by the Marine Corps.

3.2 Parts and materials. Parts and materials shall be as specified herein. Parts and materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

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3.2.1 Nonstandard parts and materials. When it has been determined that requirements cannot be met by parts and materials specified herein, immediate action shall be taken by the supplier to inform the contracting officer by submission of detailed data, requesting approval of nonstandard parts and materials.

3.3 Design and construction. The lubricating and servicing unit shall consist essentially of a gasoline engine; an electric motor; an air compressor; a high pressure cleaning system; a hydraulic oil filtration system; parts washer; air-powered lubrication pumps; hoses; hose reels and accessories. The unit shall be mounted on a steel skid framework to form a self-contained unit as shown in Figure 1.

3.3.1 Gasoline engine. The gasoline engine shall be a Wisconsin model S-100, or equal.

3.3.1.1 Power and speed rating. The power and speed rating of the gasoline engine shall be such that operation of the unit at rated power requirements and conditions shall not require horsepower (hp) in excess of the continuous brake hp rating of the engine for the applicable speed as established by the engine qualification test.

3.3.1.2 Automatic idle device. The gasoline engine shall be provided with an automatic control to reduce the engine speed to the idle speed recommended by the engine manufacturer during compressor unload period. Means shall be provided to render this device inoperative and allow the engine to operate at governed speed during unload period.

3.3.2 Fuel system. The fuel system shall be provided with a three-way selector valve conforming to MS35392-1. The auxiliary connection of the selector valve shall be provided with a plug. A corrosion-resistant circular faceplate shall be mounted on the fuel selector valve and shall be marked "OFF", "UNIT TANK", and "AUXILIARY". A fuel tank having a capacity of not less than 10 hours continuous operation shall be provided. The fuel tank shall be mounted in a manner to prevent fuel from spilling on the engine during refueling. A standpipe not less than 1/2 inch high from the bottom shall be incorporated in the fuel tank to prevent entry of dirt and foreign matter into the system. The tank shall be equipped with a drain line and shut-off valve to completely drain the tank to the outside of the unit.

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ITEM	PARTS LIST nomenclature	QTY.
1	SKID	1
2	COMPRESSOR	1
3	OIL FILTRATION SYSTEM	1
4	HIGH PRESSURE CLEANING SYSTEM	1
5	PARTS WASHER	1
6	55 GAL. DRUM	3
7	TIE DOWNS	3
8	GREASE DRUM	1
9	A.T.F. PUMP	1
10	GEAR OIL PUMP	1
11	MOTOR OIL PUMP	1
12	GREASE PUMP	1
13	AIR REEL	1
14	GREASE REEL	1
15	GEAR OIL REEL	1
16	MOTOR OIL REEL	1
17	A.T.F. REEL	1
18	REEL CABINET	1
19	UTILITY PUMP	1
20	SPARE LIGHTS	4
21	TROUBLE LIGHT	1
22	SPARE SHOE	4
23	DAVIT HOIST	1
24	TOOL CABINET	1
25	SUCTION STRAINER ASSY.	1
26	SKID STRIP	1

See Figure 2

See Figure 3

See Figure 4

Figure 1. Assembly of Lubricating and Servicing Unit (Top and Right Hand Views)

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3.3.3 Engine-exhaust system. The engine-exhaust system shall be made of insulated flexible tubing, rigid tubing, and a diverter. The insulation shall not peel, flake, crack, or otherwise break loose from the exhaust lines. The diverter shall deflect the exhaust gases either through the grease or cleaning unit as desired and into the atmosphere. The opening for the discharge to the atmosphere shall be located so as not to create a hazard. The tubing shall be of high temperature (aluminized) steel or stainless steel conforming to ASTM A268. Flexible tubing shall be used where necessary to facilitate installation and maintenance. The back pressure of the exhaust system shall not exceed 8 inches of water at any load up to the net continuous hp output and speed under all operating conditions specified herein.

3.3.4 Cranking system. Both manual and electric cranking systems conforming to MIL-E-11275 shall be provided. The manual cranking system shall be readily accessible for starting the engine.

3.3.5 Generator. The battery-changing generator shall be of a 5-ampere rating conforming to MS35876, and shall be powered by the drive shaft.

3.3.6 Batteries. Two storage batteries conforming to MS35000-1, type 2Hn shall be provided. The batteries shall be connected in series to produce 24 volts (V). The batteries shall be furnished charged and dry with a separately packaged filling of electrolyte for each unit.

3.3.7 Battery cable and terminals. The battery cables shall conform to MIL-C-13486, type I, class A and the battery terminals shall conform to MIL-E-11275. The cables shall be of sufficient length to permit the battery drawer to be pulled out and the batteries serviced without disconnecting the cables.

3.3.8 Power take-off unit. A power take-off unit shall be provided and shall consist of a clutch, housing, shaft, bearings and all accessories necessary to transmit maximum engine brake hp to the drive shaft.

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3.3.9 Coupling. The coupling shall transmit power from the power take-off to the drive shaft through an arrangement which will reduce the transmission of torsional vibration. The engine alignment shall fall within the operating tolerance of the coupling used. The coupling shall be the dry, flexible type, providing for angular and parallel misalignment.

3.3.10 Drive shaft. A drive shaft shall be provided. The shaft shall be capable of transmitting the maximum engine or motor hp without failure or permanent deformation to the air compressor.

3.3.11 Electric motor. The electric motor shall be a 7-1/2 hp, 308 V, 3 phase, 60 hertz totally enclosed type. The motor shall be furnished with a physical-disconnect type connection to a dry, flexible type coupling which is connected to the drive shaft.

3.3.11.1 Control. A combination "OFF-ON" switch and line starter with 3-phase protection shall be mounted on the unit with the necessary wiring for control of the electric motor. The line starter shall be in a splashproof, weatherproof, sleetproof, and moisture-resistant enclosure.

3.3.12 Air compressor. The air compressor shall be of the 2-cylinder, 2-stage, air-cooled type. Cooling fins shall be provided on the cylinder and cylinder heads to provide uniform cooling. The cylinder heads or cylinder assemblies, if head and cylinders are cast integrally, shall be removable. The intake and exhaust valves shall be removable. The cooling fan shall be integral with the drive pulley.

3.3.12.1 Capacity. The air compressor shall have a free air capacity of 15 cubic feet per minute (ft^3/min) at a rated discharge pressure of 175 pounds-force per square inch (lb_f/in^2) gage. Maximum discharge pressure shall be 200 lb_f/in^2 gage. The air compressor shall produce rated capacity for 500 hours at rated discharge pressure. Air compressor speed shall not exceed the manufacturer's recommended continuous operating speed.

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3.3.12.2 Intercooler. An intercooler shall be provided to cool the air between the high- and low-pressure stages, and shall be located to receive the air blast from the drive pulley fan. The discharge air temperature, measured in the discharge to the air receiver, shall not exceed the ambient temperature +100°F at a pressure of 175 +10 lb_f/in^2 gage.

3.3.12.3 Intake filter and muffler. The air compressor shall be furnished with a combination or separate intake filter and muffler. The filter shall be of the dry type, and shall remove dust and sand particles 50 microns nominal size and larger from the intake air.

3.3.12.4 Pressure regulation. An adjustable pressure-actuated mechanical control shall be provided to stop the compression of air when the receiver tank pressure has reached 175 ±10 lb_f/in^2 gage, and shall start the compression of air when the tank pressure has dropped to 140 ±10 lb_f/in^2 gage. An automatic unloading device shall be provided to protect the engine from starting against either tank pressure or compression in the compressor cylinders. A spring-loaded pressure relief valve shall be installed between the high- and low-pressure stages to protect the low-pressure cylinder in the event of failure of the high-pressure cylinder valves.

3.3.12.5 Air-receiver tank. The air-receiver tank shall be of the horizontal type and shall be constructed in accordance with the ASME Boiler and Pressure Vessel Code for Unfired Pressure Vessels. The receiver tank shall have a capacity of 30 gallons and shall be fitted with an adjustable, spring-loaded, pressure-relief valve, a female NPT globe-type 250 lb_f/in^2 gage discharge valve and drain valve, and an air line to a 0 to 250 lb_f/in^2 gage range pressure gage. The pressure gage shall be located in the control and instrument panel (see 3.3.2.5). The relief valve shall be set at 200 lb_f/in^2 gage and shall be capable of relieving the capacity of the air compressor. The takeoff point to the 1/2-inch discharge valve shall be located on the top or side of the tank, while the takeoff point for the 1/4-inch drain valve shall be located on the bottom of the tank. The drain valve shall be located at the rear of the unit, properly supported in the panel below the hose reels, and shall completely drain the tank of all moisture.

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3.3.12.6 Drive. The air compressor shall be driven by the engine or motor specified herein through matched multiple V-belts and pulleys on the drive shaft and air compressor. V-belt drives shall be in accordance with the RMA Engineering Standards for Multiple V-Belt Drives.

3.3.12.7 Mounting. The air compressor and engine/motor assembly shall be tank-mounted.

3.3.13 Parts washer. The unit shall include a parts cleaner tank which shall have a minimum capacity of 15 gallons of solvent. The tank shall be constructed of stainless steel. The tank shall contain a parts wash basket, 18 inches by 22 inches, with provisions for racking the basket for draining. The tank shall have provisions for draining completely. An access lid opening from the top is required. Seals and latches shall be provided to prevent solvent from escaping when the unit is being transported.

3.3.14 High pressure cleaning system. The high pressure cleaning system shall be powered by a 4-1/2-inch minimum air motor with a 10:1 minimum pressure ratio. All wetted parts of this system shall be constructed of stainless steel (with exception of hoses, seals, etc.) The water and cleaning solution reservoir shall have a minimum capacity of 37 gallons. The reservoir shall be constructed of stainless steel. An air pressure regulator which is compatible with the working pressures shall be included with the high pressure cleaning system. The high pressure cleaning system hose reel assembly shall consist of a corrosion resistant solution control valve, 10-inch extension tube from valve to nozzle, solution nozzle with 40° (fan width) 0.062 (equivalent orifice) stainless steel spray tip, heavy duty spring return hose reel, and 50 feet of high pressure hose included on the reel.

3.3.15 Hydraulic oil filtration system. The hydraulic oil filtration system (see Figure 2) shall consist of a minimum 50-gallon reservoir for clean "make-up" oil, suction and return hoses mounted on heavy duty rewind reels, water and dirt filter units and elements, air operated pump and necessary valving to pump oil from the vehicle hydraulic

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reservoir through the filter to "make-up" tank, or through the return hose reel to the vehicle hydraulic reservoir. Valving shall also be provided to pump from "make-up" tank through filter, to the return hose reel to the vehicle hydraulic reservoir. The filter element shall be capable of filtering a minimum of 15 gallons of oil per minute and retaining up to 1 quart of water, and shall be capable of removing 5-micron particles. A color coded differential pressure gage shall be included to indicate filter contamination. The suction hose shall be a minimum of 12 feet long and shall have a maximum outside diameter of 1-1/2 inches. The suction tube shall conform to QQ-A-200/9, 1-inch schedule, 40 aluminum pipe approximately 16 inches long. The suction hose shall be mounted on a heavy duty spring return type hose reel assembly. The hydraulic oil return hose reel assembly shall include a hose with a minimum length of 50 feet, a control valve and a heavy duty spring type return reel assembly. The reservoir shall be constructed of carbon steel conforming to QQ-S-741.

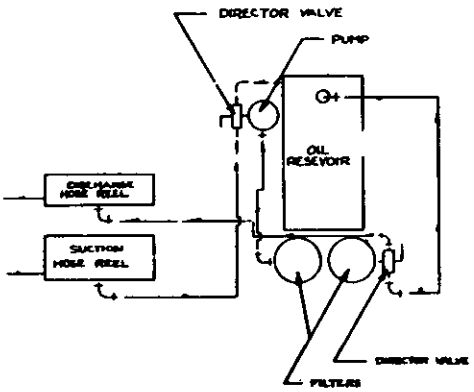


Figure 2. Hydraulic Oil Filtration System.

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3.3.16 Lubricant reservoirs. The lubricant reservoirs shall consist of three 55-gallon drums conforming to PPP-D-729, type I for gear oil, motor oil and automatic transmission fluid respectively and a 120-pound drum conforming to PPP-D-705, type III for grease.

3.3.17 Drum tiedowns. The 55-gallon drum tiedowns shall consist of four holddown rods and fittings for each drum. The rods shall be permanently secured from a base fitting which can be an integral part of the skid assembly.

3.3.18 Drum mounting. The 120-pound grease drum shall be mounted on a base of hollow sheet metal conforming to QQ-S-700. The engine exhaust shall be directed through this base to preheat the grease. An air operated elevator shall be included with the base assembly to mount the air powered grease pump. The height of the elevator in the "down" position shall not exceed 46 inches.

3.3.19 Lubricant pumps. The lubricant pumps shall be reciprocating, air-powered, heavy duty industrial type. Each pump shall operate at an air pressure of 120 lb_f/in^2 gage and shall withstand air pressure up to 200 lb_f/in^2 gage. Each pump shall be furnished with an air-pressure regulator and an air-pressure gage ranging from 0 to 200 lb_f/in^2. One master valve shall be furnished and installed to relieve the air pressure in all the pump lines. The weight of each pump with air-pressure regulator and gage shall be not more than 75 pounds.

3.3.19.1 Automatic transmission fluid pump. The automatic transmission fluid pump shall have a minimum 3-inch air motor and 5:1 pressure ratio. The pump shall transfer automatic transmission fluid from the 55-gallon drum through a hose-reel assembly to a control valve. The pump shall be of a short overall length and shall be mounted near the center of the skid assembly. A steel frame shall be provided around the pump to protect it during removal and installation of drums on the skid. An air pressure regulator shall be provided to control the air supply to the pump. A siphon hose-tube assembly shall be provided with the pump. The hose shall be a minimum of 6 feet in length. The siphon tube shall fit into the 2-inch bung opening of the drum and shall reach to the bottom of the drum. A watertight adapter shall be included to prevent water from entering through the opening.

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3.3.19.2 Gear lubricant pump. The gear lubricant pump shall have a minimum 3-inch air motor and 5:1 pressure ratio. Installation and accessories shall be identical to those in 3.3.19.1.

3.3.19.3 Motor oil pump. The motor oil pump shall have a minimum 3-inch air motor and 5:1 pressure ratio. Installation and accessories shall be identical to those in 3.3.19.1.

3.3.19.4 Chassis lubricant (grease) pump. The chassis lubricant pump shall have a minimum 3-inch air motor and 45:1 pressure ratio. The pump shall be mounted on an air powered elevator which shall raise and lower the pump into the 120-pound grease drum. Accessories shall include an air pressure regulator and a follower or induction type plate which prevents "channelling" from occurring in the grease container. A cover shall be provided to prevent contamination of the grease.

3.3.19.5 Utility transfer pump. A utility transfer pump shall be provided as an accessory to the unit. This pump shall be stored in the tool cabinet. The pump shall be equipped with an air flow control valve and quick coupler stud which will connect to the air service reel quick coupler socket. The pressure ratio shall be 1:1 and the pump must deliver 6 ounces per cycle. The pump shall fit the 2-inch opening in a 55-gallon drum and the intake shall extend to the bottom of the drum.

3.3.20 Alcohol injector. The main air line to the lubricant pumps shall be provided with an automatic alcohol injector having a metal bowl of not less than 8-fluid-ounce capacity. The control valve on the injector shall regulate the flow so that in 45 minutes of lubricating operations not more than one bowl full of alcohol shall be dispensed. Instructions for setting the control valve so the device dispenses 9 fluid ounces per hour for varying conditions of operation shall be set forth in the operator's manual and indicated on the dispenser. The fill cap or plug shall be operable by personnel wearing arctic mittens, and shall be attached to the dispenser by sash chain.

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3.3.21 Davit drum hoist assembly. A removable davit drum hoist assembly (see Figure 3), shall be provided to facilitate removal and installation of the drums. When in use, the hoist shall be supported in sockets located at different locations on the skid to allow service to all drums. The assembly shall include a minimum 1,000-pound capacity winch with 30:1 mechanical advantage, winch support with approximate overall height of 85 inches and swing radius of 30 inches, necessary pulleys and drum grabs. Provisions shall be furnished for storing the davit for transport.

3.3.22 Hose-reel assemblies. Hose-reel assemblies shall be provided as follows.

3.3.22.1 Air hose-reel assembly. The air hose-reel assembly shall be of heavy duty design, mounted on anti-friction bearings, automatically retractable, with a jam-proof ratchet, latch, and clutch and shall be installed near the right front corner of the unit. Included on the air hose-reel assembly shall be 50 feet of 3/8-inch inside diameter rayon braid air hose with a quick coupler.

3.3.22.2 High pressure chassis lubricant hose-reel assembly. The high pressure chassis lubricant hose-reel assembly shall be of heavy duty design, mounted on antifriction bearings, automatically retractable, with a jam-proof ratchet, latch, and clutch and shall be part of a reel bank located at the rear of the unit. Included on this assembly shall be 50 feet of 3/8-inch inside diameter, 2-wire braid, high pressure hose. This assembly shall include a "Z" swivel, high pressure control valve, hydraulic 12-inch whip end hose and hydraulic grease fitting coupler.

3.3.22.3 Gear lubricant hose-reel assembly. The gear lubricant hose-reel assembly shall be of heavy duty design, mounted on antifriction bearings, automatically retractable, with a jam-proof ratchet, latch, and clutch and shall be part of a reel bank located at the rear of the unit. Included on this assembly shall be 50 feet of 3/8-inch inside diameter 2-wire braid hose. This assembly shall include an 8-pint nontotalizing meter control valve with an open lube nozzle.

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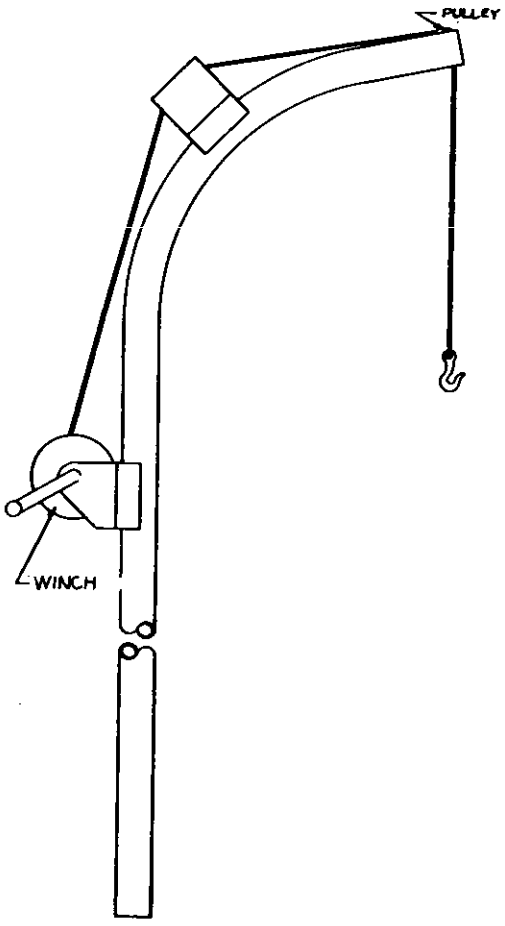


Figure 3. Davit Drum Hoist Assembly

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3.3.22.4 Motor oil hose reel assembly. The motor oil hose reel assembly shall be of heavy duty design, mounted on antifriction bearings, automatically retractable, with a jam-proof ratchet, latch, and clutch and shall be part of a reel bank located at the rear of the unit. Included on this assembly shall be 50 feet of 3/8-inch inside diameter 2-wire braid hose with a control valve incorporating a 16-quart totalizing meter and nondrip nozzle.

3.3.22.5 Automatic transmission fluid hose reel assembly. The automatic transmission fluid hose reel assembly shall be of heavy duty design, mounted on antifriction bearings, automatically retractable, with a jam-proof ratchet, latch, and clutch and shall be part of a reel bank located at the rear of the unit. Included on this assembly shall be 50 feet of 3/8-inch inside diameter 2-wire braid hose with a control valve, incorporating a 16-quart totalizing meter and nondrip nozzle.

3.3.23 Reel bank. The hose reel assemblies in 3.3.22 and the two in the hydraulic oil filtration system in 3.3.15 shall be mounted in a reel bank. The reel bank shall be located above the tool box storage compartment by means of a suitable holding device. The reels shall accommodate the hose-reel assemblies specified in 3.3.22.1 through 3.3.22.5 when the hose in each assembly is reeled in without benefit of guiding. Each reel stand shall be furnished with a brake or locking device to hold the reel in rigid position when not in use. Reel shafts and connections shall have packing to prevent leaking other than that leakage required for lubrication of the reel. Elbow unions shall be provided at the inlet side of the shaft and discharge side of the shaft to facilitate hose removal. Openings fitted with rollers to guide each hose shall be provided in a panel below the reels. Each hose reel control valve shall be mounted above the tool box storage compartment by means of a suitable holding device.

3.3.24 Skid assembly. The skid assembly shall be constructed of structural steel conforming to QQ-S-741. The assembly shall be fabricated in a manner to permit the installation of ducting for the exhaust to be used for heating of the 120-pound grease drum and the 37-gallon

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cleaner tank. The skid shall be welded into a rigid support for the unit. The skid shall also include two adjustable stops on each side to prevent the skid from sliding back and forth in the truck or trailer bed (see Figure 4). The surface of each of these stops which contacts the side of the truck or trailer shall have an area of not less than 20 square inches. These stops shall make the total overall width of the skid adjustable from 82 inches to 89 inches. The fully loaded "ready to lube" unit shall be capable of being handled by lifting attachments without bending. Lifting attachment shall consist of four lifting rings located near each corner of the skid, approximately 36 inches from the base.

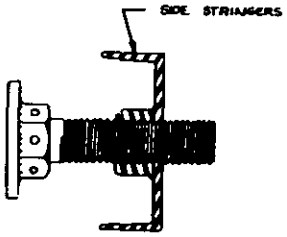


Figure 4. Adjustable Skid Stop

3.3.25 Control and instrument panel. A control and instrument panel shall be provided, mounted so as to be accessible. The panel shall include a starter switch, choke control, manual throttle control, ignition switch, and the following instruments:

(a) Battery-generator indicating ammeter and shunt, with zero center scale, conforming to MIL-M-10304.

(b) Engine-oil-pressure indicator conforming to MS24541. The oil-pressure transmitter shall conform to MS24538.

(c) Engine-fuel-liquid indicator conforming to MS24544.

(d) Air receiver pressure gage, 0 to 250 lb_f/in^2 gage range, with increments no greater than 10 lb_f/in^2.

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Each instrument and control device shall be permanently and legibly marked on plates conforming to MIL-P-514, type III, composition C, of type I, grade A, class 1 material to indicate its proper function. Not less than two panel lights with an on-off switch shall be provided to illuminate all controls and instruments.

3.3.26 Service lighting. Four 24-V sealed beam lights shall be installed on telescoping stanchions, two at the rear of the unit, and two at the front of the unit. The lights shall be provided with wiring suitable for plugging into a 24-V trailer-truck electrical system and the 24-V electrical system of the unit. The lights shall provide "flood" type lighting.

3.3.27 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating. The unit shall operate as specified herein when lubricated with military lubricants.

3.3.27.1 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I, II, or III. Fittings shall be located in a protected position. Fittings shall be accessible to a grease gun conforming to MIL-G-3859. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts.

3.3.27.2 Enclosures with integral reservoir of lubricant. Enclosures such as gear cases and transmission housings which contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, finger holes, or sight holes to determine the level of the lubricant. Each enclosure shall be equipped with a magnetic drain plug conforming to MS49006 and a means for filling the enclosure with lubricant. The drain plug shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be to the ground when the unit is in its operating position. Integral tubes or troughs may be used to convey the lubricant from the drain to the ground. Accessibility to the drain plug, the filling means, and the lubricant-level checking device shall be provided without the removal or adjustment of accessories or parts.

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3.3.27.3 Grease lubrication. When delivered, all components except those for which preservatives are specified in 5.2 shall be lubricated. All grease lubrication of sealed bearings shall be with grease as follows:

(a) MIL-G-23827 - For instruments, all sealed bearings, and other bearings where operating temperatures do not exceed 250°F.

(b) MIL-G-10924 - For bearings where operating temperatures do not exceed 175°F.

The unit shall be assembled, tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases are not always compatible with other greases. The lubricated component shall be tagged in a conspicuous place to indicate which military grease has been used.

3.3.27.4 Oil lubrication. Unless otherwise specified herein, all components shall be filled to the operating level with military oils designated for use in the temperature range at the shipping destination. A tag shall be attached in a conspicuous place to indicate the temperature range and grade of oil used.

3.3.28 Retractable trouble light-reel assembly. A retractable trouble light-reel assembly shall be provided. This assembly shall be located at the rear of the unit and shall consist of a heavy-duty type spring retractable reel, 50 feet of heavy-duty waterproof electrical cord, trouble light socket (weatherproof) and a 24-V direct current (dc). bulb. The source of power shall be the 24-V electrical system of the unit.

3.3.29 Tool and accessory storage cabinet. A storage cabinet shall be provided under the hose outlet valves to provide storage for eight extra hydraulic oil filter cartridges, tools and accessories. The top of the storage cabinet shall be designed as a drip pan for the lubricant control valves. The tool storage area shall be to the right of the filter storage area. The tool storage area

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shall provide storage space for hand tools and accessories in 3.3.31. All doors to the storage cabinet shall be of a dustproof/waterproof design. Brackets shall be provided to secure utility transfer pump in the storage cabinet. Brackets shall also be provided to secure miscellaneous accessories in storage cabinet. Small parts drawers shall be provided in tool storage cabinet. A drawer shall also be provided for lube charts 9 inches by 12 inches by 1-1/2 inches thick.

3.3.29.1 Dustproof/waterproof control valve cover. A cover shall be provided at the top of the storage cabinet to prevent excessive contamination of the drip pan and control valves. All access doors shall have suitable seals to provide dustproofing/waterproofing. The back and sides shall be enclosed.

3.3.30 Battery drawer. A metal battery drawer to hold the storage batteries shall be provided, located under the hose-reel bank. The drawer shall incorporate provisions for gas venting, drainage, rigid battery support on non-conductive shims, corrosion-resistant battery holddown devices, polychloroprene pans, and access to battery cells and terminals from the rear of the unit. Shims and spacers shall not interfere with the free circulation of air between batteries and drawer. Shims shall be attached to the drawer bottom with machine screws or bolts with nuts and lock washers. A device shall be provided to automatically secure the drawer in place when it is closed. The drawer, with batteries and cables installed, shall require not more than a 40-pound force to slide it in and out. A stop shall be provided to limit outward travel of the drawer.

3.3.31 Accessories. Accessories shall be provided as follows:

Item	Quantity	Description
1	2	Adapter, flexible extension, locking sleeve on one end and hydraulic coupling on the other conforming to MIL-L-4387, type IV, class 1, style B, except that the hose shall conform to MIL-H-3868, type 1, grade A, size 1/4 inch, and the hydraulic coupling shall be size 2 without ball-check valve.

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Item	Quantity	Description
2	1	Adapter, flexible extension as above except the hydraulic coupling shall be replaced with a large button-head coupling conforming to MIL-L-4387, type 1, class 3. The large button-head coupling shall be without check valve and fitted with a 500-lb_f/in^2 gage pressure-relief valve.
3	1	Adapter, flexible extension as above except the large button-head coupling shall be replaced with a standard button-head coupling conforming to MIL-L-4387, type 1, class 3, style B, size 1, and fitted with a 500-lb_f/in^2 gage pressure-relief valve without check valve.
4	1	Adapter, rigid, with relief, consisting of a hydraulic coupling conforming to MIL-L-4387, type 1, class 1, style A, size 2, a locking sleeve conforming to MIL-L-4387, type II, a 500-lb_f/in^2 gage pressure-relief valve and interconnecting short nipples of 1/8-inch extra heavy pipe.
5	1	Adapter, rigid, swivel coupling, conforming to MIL-L-4387, type IV, class 1, style A. The hydraulic coupling shall be without ball-check valve.
6	1	Air blowgun, conforming to GGG-G-770, type II, class 2, style A, size 1/4-inch NPT, and a quick-disconnect coupling half, conforming to MIL-C-4109, type II, style 1, size 1/4 inch, shall be installed in the inlet port.
7	1	Fluid suction gun, plunger-operated, conforming to GGG-O-591, type II, class C, style 4, 11-ounce capacity.

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Item	Quantity	Description
8	2	Grease gun, hand, lever-operated, conforming to MIL-G-3859, type I, 14-ounce capacity.
9	1	Hacksaw frame, conforming to GGG-F-671, type I, class 1, style B, 10- to 12-inch capacity.
10	12	Hacksaw blade, conforming to GGG-B-451, type I, grade A, class 2, 10 inch length, 24 teeth per inch.
11	3	Lubrication fitting, giant button head, 1/8-inch NPT.
12	3	Lubrication fitting, giant botton head, 1/4-inch NPT.
13	3	Lubrication fitting, giant button head, 3/8-inch NPT.
14	3	Lubrication fitting, standard button head, 1/8-inch NPT.
15	3	Lubrication fitting, standard button head, 1/4-inch NPT.
16	3	Lubrication fitting, standard button head, 3/8-inch NPT.
17	6	Lubrication fitting, hydraulic, straight, 1/8-inch NPT, conforming to MIL-F-3541, type III.
18	6	Lubrication fitting, hydraulic, 45 degrees, 1/8-inch NPT, conforming to MIL-F-3541, type III.
19	6	Lubrication fitting, hydraulic, straight, 1/4-28 taper threads, conforming to MIL-F-3541, type I.

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Item	Quantity	Description
20	6	Lubrication fitting, hydraulic, 45 degrees, 1/4-28 taper threads, conforming to MIL-F-3541, type I.
21	2	Lubrication fitting tool, 3 in 1, consisting of tap, wrench, and easy-out, size 1/8-inch NPT.
22	2	Lubrication fitting tool, 3 in 1, consisting of tap, wrench, and easy-out, size 1/4-28 taper threads.
23	2	Lubricator, portable, with harness for mounting on back; 5-pound grease capacity, 3,500 lb_f/in^2 gage delivered pressure, 40 lb_f/in^2 gage permanent air charge.
24	1	Oil spray gun, air-operated, 1-quart capacity, with quick-disconnect coupling half conforming to MIL-C-4109 type II, style I, size 1/4 inch.
25	2	Oiler, hand, pump feed 1/2-pint capacity, 6-inch rigid bent spout, conforming to GGG-O-591, type II, class B.
26	12	Plugs, to replace 500 lb_f/in^2 gage pressure-relief valve specified above.
27	1	Quick-disconnect coupling half (socket end with valve), conforming to MIL-C-4109, type I, style 2, size 1/4 inch.
28	3	Quick-disconnect coupling half (plug end straight through), conforming to MIL-C-4109, type II, style k, size 1/4 inch.
29	6	Reusable coupling, male, 3/8-inch NPTF, to repair lubricant hose.

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Item	Quantity	Description
30	6	Reusable coupling, female swivel, 3/8-inch NPTF, to repair lubricant hose.
31	3	Reusable coupling, male, 1/4-inch NPTF, to repair air hose.
32	3	Reusable coupling, female swivel, 1/4-inch NPTF, to repair air hose.
33	1	Swivel, straight, 360 degrees, conforming to MIL-L-4387, type VII, class 1, style A or B (as used on oil control valves.)
34	1	Swivel, "Z", 360 degrees, conforming to MIL-L-4387, type VII, class 3. (As used on grease control valves).
35	1	Tire inflator and pressure gage, chucks for standard and large-bore tire valves chained to gage, conforming to GG-G-91, type III, style 3, except that the hose shall be as specified herein and a quick-disconnect coupling half shall be attached to the air inlet of the gage. All quick-disconnect couplings shall conform to MIL-C-4109; the plugs shall be type II, style 1, size 1/4 inch NPT while the socket shall be type I, style 2, size 1/4 inch NPT.
36	1	35-lb grease pail follower plate (to be used with grease pump assembly).
37	1	15-lb carbon dioxide fire extinguisher.
38	1	30-lb dry chemical fire extinguisher.
39	1	Universal drum wrench (55-gallon drum).

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3.3.32 Steel fabrication. Flame cutting shall not be performed on stressed sections fabricated from low-alloy steel unless followed by machining of the flame-cut edge to a depth of one-quarter inch; flame cutting without machining shall be permitted if a post-heating torch is used during the cutting operation. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly. All burrs and rough edges shall be removed. All bends shall be made to a radius which shall not cause cracking at the extreme outer fibers of the piece being bent, and all bends except for minor details shall be made by use of metal dies or fixtures to insure uniformity in size and shape.

3.3.33 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loadings. Welding shall be done by the shielded electric-arc method in accordance with applicable requirements of the design, construction, and repair section of AWS Welded Highway and Railway Bridges. Where embrittlement of the joined metal may result, preheating or annealing, or both, shall be employed. All welded parts shall be free from cracks and other imperfections which may reduce the effectiveness of the part. Work shall be positioned for flat welding whenever practicable.

3.3.34 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed, and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society.

Welding Qualifications of the ASME.

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3.3.35 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Bolt holes in structural members subject to variable loads shall be punched or drilled. Washers or lockwashers shall be provided where necessary. All bolts, nuts, and screws shall be correctly torqued. Matching thread areas securing bolts or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt.

3.3.36 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member and shall be in accordance with SAE J492.

3.3.37 Threads. All screw threads shall be in accordance with H28.

3.3.38 Safety. All exposed parts subject to high operating temperatures or energized electrically shall be insulated, fully enclosed, or guarded. All moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. Protective devices shall not impair the operating functions.

3.3.38.1 Safety devices. Safety devices of the emergency shutdown type shall be provided for high oil temperature and low oil pressure. A combination ignition-start switch which incorporates the aforementioned pressure-temperature safety devices may be used in lieu of four separate switches. Safety switches shall be set in accordance with MIL-E-46717 or MIL-E-11275, as applicable.

3.3.39 Fungus and moisture resistance. Unless otherwise specified herein, the electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, except that 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish shall be used in lieu of salicylanilide, the specified fungitoxicant:

(a) Components or circuit elements which are inherently fungus and moinsture resistant or which are hermetically sealed need not be treated.

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(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherance or function of the varnish.

3.3.40 Technical publications. Such technical publications as are specified in contract or order shall be furnished (see 6.2).

3.3.41 Repair parts and tools. The contractor shall furnish a recommended listing of spare parts and tools required (see 6.2). No shipment of units shall be made unless repair parts and maintenance tools are included with the shipment or unless approval for shipment of the pumping assemblies without such items has been received from the contracting officer.

3.4 Maintainability. All components and accessories shall be reasonably accessible for maintenance. The repair, replacement and adjustment of components and accessories shall be accomplished by using common tools and with minimum drainage of and minimum disturbance to other components and accessories. Drain outlets shall be located for accessibility.

3.5 Performance. The pump performance shall be as follows:

	$-25^{\circ}F$	$+75^{\circ}F \pm 10^{\circ}F$	$+125^{\circ}F$
Chassis lubricant (grease) pump	10 ounces within 5 minutes of continuous operation	3.6 ounces in 10 pumping cycles	3.6 ounces in 10 pumping cycles
Minimum static pressure	3700 lb_f/in^2	4500 lb_f/in^2	4500 lb_f/in^2

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	-25°F	+75°F ±10°F	+125°F
Gear lubricant pump	1 gallon within 1 minute of continuous operation	3 pints within 10 pumping cycles.	3 pints within 10 pumping cycles.
Minimum static pressure	300 lb_f/in^2	300 lb_f/in^2	300 lb_f/in^2
Motor oil pump & automatic transmission fluid pump	1 gallon within 1 minute of continuous operation	3 pints within 10 pumping cycles	3 pints within 10 pumping cycles
Minimum static pressure	300 lb_f/in^2	300 lb_f/in^2	300 lb_f/in^2
Utility transfer pump	pump shall operate	12 gallons within 1 minute	12 gallons within 1 minute

NOTE: All operating air pressures to be 120 lb_f/in^2

3.6 Dimensions. The overall dimensions of the unit, including skids, shall be not more than 145 inches in length by 80 inches in width by 46 inches in height. The height of the detachable drum hoist shall not exceed 81 inches during installation and removal of drums.

3.7 Weight. The gross weight of the unit fully loaded shall not exceed 6,500 pounds.

3.8 Treatment and painting. Except as otherwise specified herein, all parts of the unit normally painted in commercial practice shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. The color shall match Marine Corps green, No. 23, semigloss (see 6.3).

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3.8.1 Battery drawer and tool and accessory storage cabinet. The interior surfaces of the battery drawer and tool and accessory storage cabinet shall be treated in accordance with MIL-T-704, and coated with epoxy conforming to MIL-C-4556; the color shall be white.

3.9 Marking. The unit shall be identified in accordance with MIL-STD-130. Identification plates shall conform to MIL-P-514, photosensitive, anodized aluminum.

3.9.1 Stenciling. The gross weight of the unit shall be stenciled on both sides of the unit in such a position as to be readily discernible. The letters shall be gothic capitals and the numerals shall be of similar appearance. The letters and numerals shall be 2 inches high where space permits.

3.9.2 Instruction plates. Each unit shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the pumping assembly. Instruction plates shall conform to MIL-P-514, photosensitive, anodized aluminum.

3.9.3 "No smoking-flammable liquid" signs. Four "No smoking-flammable liquid" signs shall be provided on each side of the unit. The letters shall be red on a white background and a minimum of 2 inches high.

3.10 Workmanship. Workmanship shall be in accordance with the accepted commercial practice for this type of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards and as specified herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).

(b) Quality conformance inspection (see 4.4).

(c) Preparation for delivery (see 4.7).

4.3 First article inspection. The first article shall be examined for defects listed in 4.5. Presence of one or more defects shall be cause for rejection. The first article shall be subjected to tests in 4.6. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection. Each unit shall be examined for defects listed in 4.5. Presence of one or more defects shall be cause for rejection. Each unit shall be subjected to test in 4.6.1. Failure to pass the test shall be cause for rejection.

4.5 Defects. The unit shall be examined for the following defects:

101. Material not as specified.
102. Safety devices or guards missing or inadequate, or sharp corners and sharp edges not rounded off.
103. Components or accessories not accessible for repair, replacement or adjustment.
104. Fasteners inadequate or not as specified.
105. Dimensions and weight not as specified.
106. Enclosure not as specified.
107. Lubricant containers not as specified.
108. Fuel tank not as specified.
109. Engine exhaust system not as specified.
110. Manual starting device not accessible.
111. Battery drawer and battery cables not as specified.
112. Control and instrument panel not as specified.

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113. Hoses not as specified.
114. Wiring not as specified.
115. Accessory drawers not as specified.
116. Accessories missing or not as specified.
117. Lubrication devices not accessible.
118. Identification or other marking, operating and safety instructions missing or not as specified.
119. Treatment and painting not as specified.
120. Toolbox not as specified.
121. Workmanship not as specified.
122. Air receiver and accessories not as specified.
123. Misalignment.
124. Manuals missing or not as specified.
125. Electric motor not as specified.
126. Electric motor controls not as specified.

4.6 Tests. Unless otherwise specified herein, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The unit shall operate as specified herein without maintenance other than the supplier's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the supplier prior to test. Before performing the following tests, the unit shall be serviced with military fuels and lubricants designated for use at the ambient temperature at which the test is being conducted.

4.6.1 Initial operation. Sufficient lubricant shall be placed in the applicable containers to prime the pumps and purge all lines and control valves. The alcohol injector shall be filled with methyl alcohol and adjusted for operation at a rate of 9 ounces per hour. The engine shall be started and the compressor operated for 30 minutes. The unit shall be checked and any one of the following shall constitute failure of this test:

(a) Erratic engine operation or indications of insufficient power and speed.

(b) Inability of the compressor to produce 175 ± 10 lb_f/in^2.

(c) Evidence of oil, grease, fuel or air leaks.

(d) V-belt or clutch slippage.

(e) Malfunction of the automatic idle device.

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(f) Alcohol injector inoperative.

(g) Delivery and developed pressure of the pumps not as specified.

Upon completion of the above, the motor shall be started and the compressor operated for 30 minutes. The unit shall be checked and any one of the following shall constitute failure of this test:

(a) Indications of insufficient power and speed.

(b) Inability of the compressor to produce 175 $\pm$10 lb_f/in^2.

(c) Evidence of oil, grease, or air leaks.

(d) Malfunction of the electric controls.

(e) Delivery and developed pressure of the pumps not as specified.

(f) V-belt or clutch slippage or malfunction.

4.6.2 Low temperature. The lubricant containers shall be filled not less than 3/4 full of the lubricants for use at -25°F. Temperature probes shall be placed as close as practicable to the pump suctions, the engine crankcase, and in one of the central cells of each battery. The alcohol injector shall be filled with methyl alcohol and adjusted for operation at 9 ounces per hour. The unit shall be cooled to a temperature of -25°F. The automatic idle device shall be set in the "off" position. After all temperatures have stabilized at not more than -25°F and while maintaining the chamber temperature at not more than -25°F , the following tests shall be performed.

4.6.2.1 Accessibility. Accessibility shall be tested as follows:

(a) Close the battery drawer and then pull it open.

(b) Close and open all unit doors including container covers.

(c) Completely unreel each hose in turn and then reel it up again, checking operation of reel brakes, reels, and control-valve supports.

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(d) Open and close all valves.

(e) Remove transfer pump and connect air hose and discharge hose. Replace transfer pump and hose.

(f) Remove operating accessories from accessory drawers, connect or attach the items, if applicable, and then replace.

(g) Operate complete lighting system.

A force in excess of 40 pounds for item (a) or inability to perform any operation (a) through (g) shall constitute failure of this test.

4.6.2.2 Operational. While at -25°F., the unit shall be started and operated for 1 hour using the engine and for 1 hour using the motor. Failure of the unit to start and dispense lubricants within 15 minutes as specified in 3.5 shall constitute failure of this test.

4.6.3 Endurance. At the start of the endurance test, the following parameters shall be measured:

(a) Engine exhaust back pressure.

(b) Air compressor discharge air temperature.

(c) Motor kilowatts (kW).

Failure of the engine exhaust to be as specified in 3.3.3, motor drawing kW in excess of manufacturer's rating, or the discharge air temperature to be as specified in 3.3.12.1 shall constitute failure of this test. The unit shall be subjected to an endurance test of 250 hours actual running time with the gasoline engine and 250 hours actual running time with the electric motor at rated capacity of 15 ft^3/min and rated pressure of 175 lb_f/in^2 gage. At intervals of 3 hours operating time, with the unit running, the following operations shall be performed:

(a) Each control valve shall be operated for a minimum of 10 seconds each to recirculate lubricant into their respective containers.

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(b) Air shall be discharged through the air receiver drain valve until the air receiver pressure is 125 lb_f/in^2 gage.

Failure of the air compressor to complete 500 hours actual running time at rated capacity and pressure without slippage, overheating, breakage or other malfunctions, or failure of any other component shall constitute failure of this test.

4.6.4 High temperature. The unit shall be warmed to a temperature of +120°F. With the temperatures of the oils, greases, and battery electrolyte stabilized and while maintaining a chamber temperature of +120°F , the unit shall be started and operated for 1 hour using the engine and 1 hour using the motor. Failure of the unit to start and dispense lubricant as specified in 3.5, or failure to operate for 2 hours shall constitute failure of this test.

4.6.5 Lift. The unit shall be lifted by the lifting rings to a height of 10 feet, lowered at least 2 feet in a free fall and braked suddenly. Weld failure or permanent deformation of the unit shall constitute failure of this test.

4.6.6 Road. The unit shall be loaded on a truck which is compatible to the M35A2C truck. The unit, filled with appropriate lubricants, shall be driven at maximum safe speeds for 50 miles on unimproved roads and 50 miles on cross-country trails. The towing on unimproved roads shall include not less than 15 abrupt stops from a speed of 25 miles per hour (mi/h) in forward gear, and not less than 10 abrupt stops from a speed of 5 mi/h in reverse gear. Upon completion of the road testing, the unit shall be subjected to test in 4.6.1. Damage or displacement of any component, accessory, part or tool in or on the unit, or evidence of defective welded, bolted, or riveted connections on the unit, or failure of the adjustable stops or failure to meet the requirements of 4.6.1 shall constitute failure of this test.

4.7 Inspection of preparation for delivery.

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4.7.1 First article pack inspection. When first article pack is required (see 5.1), it shall be examined for defects listed in 4.7.3. Presence of one or more defects shall be cause for rejection.

4.7.2 Quality conformance inspection. For the purpose of this inspection, a completed pack prepared for shipment shall be considered a unit of product. Sampling for examination shall be in accordance with MIL-STD-105. Samples selected shall be examined for defects listed in 4.7.3. Acceptable quality level shall be 2.5 percent defective. A lot shall consist of all completed packs of any one level offered for examination at one time.

4.7.3 Defects. The first article pack shall be examined for the following defects:

127. Materials, methods, and containers not as specified. Each incorrect material, method, or container shall constitute one defect.
128. Coated surfaces of machined surfaces not wrapped or covered with barrier-material as specified.
129. Unpainted metal information plates not coated with varnish as specified.
130. Clutches not blocked or disengaged as specified.
131. Pulley grooves not coated with primer as specified.
132. Surfaces of pumps not coated with preservatives as specified.
133. Interior surfaces of alcohol injector not coated with preservative as specified.
134. Hoses not secured to prevent unwinding as specified.
135. Open ends of hoses not sealed with tape as specified.
136. Interior of padlocks not treated with graphite as specified.
137. Portable lubricators not coated with preservatives as specified.
138. Fire extinguishers not removed and packaged as specified.
139. Intermediate packaging not as specified.
140. Strapping on crates not zinc coated as specified.
141. Marking illegible, incomplete, or incorrect.

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4.7.4 Test. The first article pack shall be tested in accordance with MIL-STD-705, method T740.5, at speeds up to and including 10 mi/h. Any structural damage which precludes normal operation of the unit shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 First article pack. Unless otherwise specified (see 6.2), the supplier shall furnish a first article pack for examination within the time frame specified (see 6.2) to prove, prior to starting production packaging, that the preservation, packaging, packing and marking comply with the preparation for delivery requirements of this specification. Examination shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government.

5.2 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in, and shall be applied in accordance with,MIL-P-116.

5.2.1.2 Unprotected metal surfaces. Unpainted exterior unfinished metal surfaces of the unit and components requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. Exterior machined surfaces shall be coated with type P-6 preservative. The coated surfaces shall be wrapped or covered with greaseproof barrier material conforming to MIL-B-121, type II, grade A, class 2, secured in place with tape conforming to PPP-T-60, type IV, class 1.

5.2.1.3 Engines. Engines and engine accessories shall be preserved in accordance with MIL-E-10062, level A, method I, classification type II.

5.2.1.4 Batteries. Charged and dry batteries shall be placed and secured in the battery drawer. The cables shall be disconnected from the battery and secured to the battery box with tape conforming to PPP-T-60, type IV, class 1.

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The vent holes in the fill caps shall be plugged or sealed. Electrolyte shall be packed in type IV units in accordance with MIL-S-207. The packaged electrolyte shall be over-packed in wood boxes in accordance with MIL-S-207. The packed electrolyte shall be secured to the unit or to the base of the crate in a protected location.

5.2.1.5 Disk-type clutch. The inspection plate shall be removed and, with the clutch engaged, all interior components of the clutch shall be sprayed with primer conforming to TT-P-664. Clutch control mechanisms not enclosed shall be coated with preservative conforming to MIL-P-116, type P-1. After spraying, spring-loaded-type clutches shall be blocked in a partially disengaged position. Snap-over-center and toggle-in-type clutches shall be completely disengaged.

5.2.1.6 Drive belts and pulleys. Drive belts shall be removed or released from tension. Pulley grooves shall be coated with primer conforming to TT-P-664. Removed belts shall be preserved in accordance with MIL-P-116, method IC-1.

5.2.1.7 Air compressor and air receiver. The air compressor and air receiver shall be preserved in accordance with MIL-C-3600, level A.

5.2.1.8 Lubricant pumps. The interior surfaces and working parts of the pumps and the air motors except the grease pump, shall be coated with preservative conforming to MIL-P-116, type P-10, grade 1. The preservative shall be applied by actuating the pumps during the application of the preservative, and the excess preservative shall be drained. The interior surfaces of the grease pump shall be coated with preservative conforming to MIL-P-116, type P-11. Exterior surfaces requiring a preservative in accordance with MIL-P-116 shall be coated with type P-7 preservative. The pumps shall be secured on the lubricant container compartment.

5.2.1.9 Alcohol injector. The interior surfaces of the alcohol injector shall be coated with preservative conforming to MIL-P-116, type P-10, grade 1.

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5.2.1.10 Control valves. The control valves shall be individually preserved in accordance with MIL-P-116, method IC-1, applying type P-10, grade 1 preservative to the internal and external surfaces.

5.2.1.11 Hoses. Hoses attached to reels shall be compactly wound on the reels and secured to prevent unwinding, and the reels shall be locked. Hoses not attached to reels shall be coiled to a minimum safe diameter, and the coils shall be secured with evenly spaced wraps of tape conforming to PPP-T-60, type IV, class 1, or with cotton tape. The open ends of the hoses shall be sealed with tape specified above.

5.2.1.12 Electric motors. Electric motors shall be preserved in accordance with MIL-E-16298 by the alternate packaging method as specified for equipment assembled to machines.

5.2.1.13 Accessories. Accessories shall be preserved as follows.

5.2.1.13.1 Padlocks and keys. The interior of the padlocks shall be treated with powdered graphite applied through the key opening. The locks shall be operated to insure coverage of interior parts and surfaces. Each lock with keys shall then be preserved in accordance with MIL-P-116, method IC-1.

5.2.1.13.2 Lubrication fittings and adapters. Lubrication fittings and adapters of like size and description shall be preserved and packaged together in accordance with MIL-P-116, method IC-1, applying type P-7 preservative when required by the preservative application criteria of MIL-P-116.

5.2.1.13.3 Tools and lubricating guns. Tools such as the hacksaw frame and blades and the lubricating guns shall be preserved in accordance with PPP-P-40.

5.2.1.13.4 Portable lubricator. The interior surface of the lubricator shall be coated with MIL-P-116, type P-7 preservative. Exterior metal surfaces requiring a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. The lubricator shall be placed and secured in its mounting.

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5.2.1.13.5 Other accessories. Other accessories of like description shall be preserved and packaged together in accordance with MIL-P-116, method IC-1, applying type P-7 preservative to surfaces of the item requiring a contact preservative in accordance with MIL-P-116.

5.2.1.14 Fire extinguisher. Exterior metal surfaces of the fire extinguisher requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-1 preservative. The fire extinguisher shall be packaged in a box conforming to class weather resistant of PPP-B-636. The box shall be waterproof sealed with tape as specified for slotted style boxes in accordance with the appendix to the box specification.

5.2.1.15 Technical publications. Technical publications for each unit shall be preserved and packaged together in accordance with MIL-P-116, method IC-1.

5.2.1.16 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts.

5.2.1.17 Maintenance tools. Maintenance tools shall be preserved in accordance with PPP-P-40.

5.2.1.18 Intermediate packaging. Packaged and unpackaged components, repair parts, tools, publications, and accessories shall be placed in the drawers, toolbox, and tool and accessories storage cabinet. Cushioning shall be provided to prevent damage. The drawers and boxes shall be closed and secured to prevent pilferage.

5.2.2 Level C. Each complete unit, repair parts, tools, accessories, and technical publications shall be preserved and packaged to afford adequate protection against deterioration and damage from the supplier to the initial destination. The supplier may use his commercial practice provided it fulfills these requirements.

5.3 Packing. Packing shall be level A or C as specified (see 6.2).

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5.3.1 Level A. Each unit, repair parts, and accessories comprising a complete unit shall be packed in a crate conforming to MIL-C-3774, nailed assembly, with skid-type base. Blocking and anchoring shall be in accordance with the appendix to MIL-C-104. Strapping shall be zinc-coated.

5.3.2 Level C. Each unit shall have the repair parts, tools, accessories, and publications packed in the drawers and tool and accessory storage cabinet on the unit. The drawers and lids of the cabinets shall be secured to prevent pilferage. The skid-mounted unit shall be packed unboxed in a manner to insure carrier acceptance and safe delivery to destination at lowest rates in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.4 Marking. Unboxed units, packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The unit is intended for use under both normal and extreme climatic conditions for the lubrication of tractors, shovels, forklifts, graders and other types of construction equipment.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.

(b) First article time frame (see 3.1).

(c) Technical publications to be furnished (see 3.3.4).

(d) When list of repair parts and maintenance tools is not required (see 3.3.41).

(e) When a first article pack is not required (see 5.1).

(f) Time frame for first article pack (see 5.1).

(g) Applicable levels of preservation, packaging and packing required (see 5.2 and 5.3).

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6.2.2 Contract data requirements: Data conforming to Data Item Descriptions DI-E-4527, DI-E-4528, DI-M-4704, DI-R-4803, DI-T-4902 and DI-T-4904 will usually be required for delivery in connection with this specification. When so required, such data will be specified for delivery on a DD Form 1423 included in the contract.

6.3 Color. Color chips for Marine Corps green, No. 23, may be obtained from the Commandant of the Marine Corps (Code CSY-10), Headquarters U. S. Marine Corps, Washington, D. C. 20380.

6.4 First article instructions. First article testing shall be in accordance with Section 4.

Preparing activity:
Navy - MC
Project No. 4930-N137

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